

NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

Third Semester B.E. (E&C) (Credit System) Degree Examinations

April 2023

21EC303 - DIGITAL SYSTEM DESIGN WITH VERILOG

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

Unit – I

	Marks	BT*	CO*
1. a) A switching circuit has four inputs A, B, C & D. A & B are the bits of number N1; C & D are the bits of number N2. The output Y of the circuit must go High when the product $N_1 \times N_2$ is greater than or equal to 4. Find the minterm and maxterm expressions.	6	L*3	1
b) Convert $Y = AB + A\bar{C} + BC$ into canonical SOP form using any method.	4	L3	1
c) Simplify $f(A, B, C, D) = \pi(0, 3, 5, 6, 7, 8, 11, 13, 14, 15)$. d(1, 2, 10) using K-map and implement using (i) NOR gates only (ii) NAND only.	10	L3	1
2. a) Obtain the simplified expression for full-subtractor and implement using two half-subtractors.	6	L2	2
b) Implement 4-bit adder using lookahead carry principle. Compare this with ripple carry adder in terms of propagation delays.	10	L3	2
c) List all the bitwise and Boolean logical operators used in Verilog.	4	L2	2
3. a) Given $F = (A + C + D)(A + B + \bar{C})(\bar{A} + B + C)(\bar{A} + \bar{C} + D)$, list all the minterms, essential prime implicants, non-essential prime implicants, simplified SOP expression and NAND implementations.	10	L2	2
b) Starting from truth-table, write the Verilog code for full-adder using data-flow modelling and obtain its simulation wave forms with no delay. Derive all the expressions used.	10	L2	2

Unit – II

4. a) Implement the functions $f_1(A, B, C, D) = \sum m(1, 2, 4, 7, 8)$ and $f_2(A, B, C, D) = \pi M(10, 12, 13, 14)$ using a single 4:16 decoder with active low outputs.	6	L3	3
b) Implement the function $f(a, b, c, d) = \sum m(2, 4, 6, 7, 9, 10, 11, 12, 15)$ using 8:1 multiplexer.	6	L3	3
c) Design and implement 2-bit magnitude comparator circuit that compares two binary numbers A_1A_0 & B_1B_0 and gives appropriate outputs.	8	L2	3
5. a) What is a switch debouncer? Design a switch debouncer using $\bar{S}\bar{R}$ Latch.	8	L2	4
b) Starting from the truth table, obtain the logic circuit and write the Verilog dataflow code for 2X2 unsigned combinational multiplier circuit, ignoring the propagation delays.	12	L3	3
6. a) The logic diagram of a proposed JK Latch is shown in Fig. Q6(a). Analyze the circuit and draw the function table. Does it work like a normal JK latch? If not, propose an alternate circuit.			

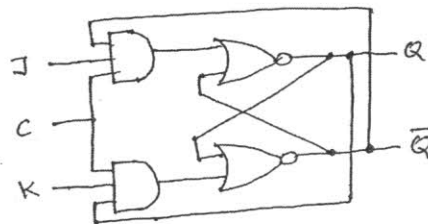


Fig. Q6(a)

P.T.O.

6 L3 4

- b) Explain 0's and 1's catching problem in JK FlipFlops. 8 L2 4
- c) Write Verilog behavioural code for positive edge triggered JK flip flop using the 'case' statement. 6 L2 4

Unit – III

7. a) Implement 4.bit binary ripple up-counter using negative edge triggered D flipflops and draw its timing diagram showing all the possible states. 6 L2 5
- b) Construct the excitation table, transition table, state table for the following excitation expressions.
 $D_A = AB + XA'B'$, $D_B = A'B' + X'A'$, $Y = XA + XB'$ 8 L3 5
- c) Write a verilog behavioral code for positive edge triggered synchronous 4-bit binary counter. 6 L3 5
8. a) Design and implement synchronous binary counter having the count sequence 4,1,7,5,2 and repeat, using D flop-flops. 8 L3 5
- b) Implement the synchronous circuit whose state diagram is shown in Fig. Q8(b), using D flip flop.

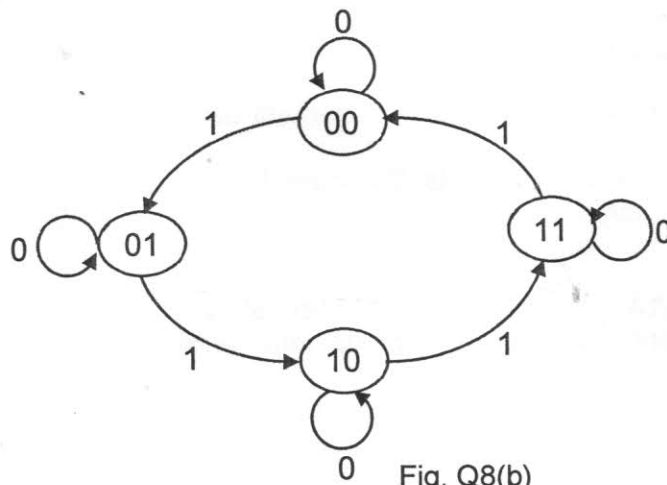


Fig. Q8(b)

- c) Write a verilog behavioral code for logical shifting of an 8-bit register. 6 L3 5

6 L3 5
6 L2 5

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome
